understanding the concept of function prototypes and scope is crucial for managing complex projects.

Function Prototype Concept:

- In Python, defining a function (`def`) implicitly declares its existence.
- The function signature, including its name and parameters, acts as the prototype.
- There's no need to declare the function separately before defining it, as in some statically-typed languages.

Example of a Function Signature:

```
def calculate_area(length: float, width: float) -> float:
    return length * width
```

- **Signature:** `calculate_area(length: float, width: float) -> float`
- This signature indicates the function's name (`calculate_area`), parameter types (both `float`), and return type (`float`).

4.3 Recursion and Its Applications

What is Recursion?

Recursion occurs when a function calls itself directly or indirectly. It's a powerful tool for solving problems that can be broken down into smaller, similar problems. Each recursive call should approach a base condition, ensuring the function doesn't call itself indefinitely.